AWS Multi-Account VPC Peering Implementation

🚀 Project Overview

This project demonstrates how to establish secure private connectivity between two Amazon VPCs, that belong to different AWS accounts using **VPC Peering**. The setup allows EC2 instances in separate AWS accounts to communicate using **private IP addresses** without routing traffic over the public internet.

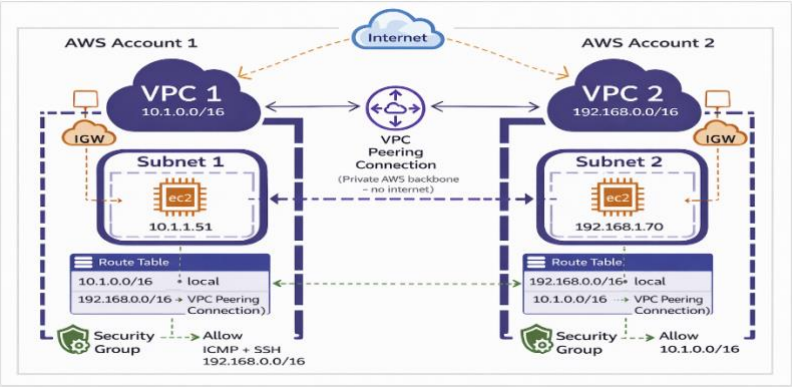
🎯 Problem Statement

VPCs across multiple AWS accounts are network-isolated by default, requiring a secure, low-latency private connectivity solution to enable inter-VPC communication without traversing the public internet.

🏗️ Workflow

Create VPCs (non-overlapping CIDRs in two AWS accounts)
↓
Launch EC2 Instances (one per VPC)
↓
Initiate VPC Peering (from requester account)
↓
Accept Peering Request (from accepter account)
↓
Update Route Tables (add peer CIDR → peering connection)
↓
Configure Security Groups (allow traffic from peer VPC CIDR)
↓
Private Traffic Flow (EC2 ↔ EC2 via AWS backbone)
↓
Validate Connectivity (ping / SSH using private IP)

Architecture Overview:



Implementation Highlights

- Implemented cross-account VPC peering with non-overlapping CIDR blocks.
- Configured route tables and security groups for private inter-VPC communication.
- Validated connectivity using ping, telnet, and SSH over private IPs.

Tools & Technologies

- Amazon VPC
- VPC Peering
- Amazon EC2
- Route Tables
- Security Groups
- Internet Gateway

Key Outcomes

- Enabled secure private communication between EC2 instances across AWS accounts.
- Achieved low-latency connectivity using AWS private backbone.
- Eliminated dependency on public internet for inter-VPC traffic.
- Validated end-to-end connectivity using private IPs (ping and SSH).

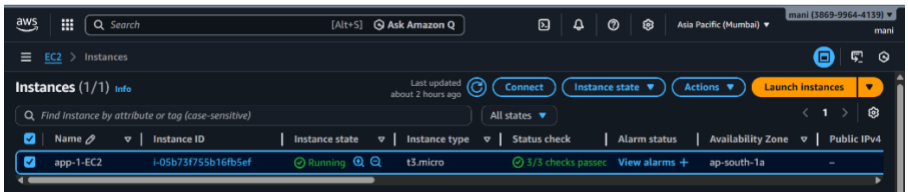
Steps to follow

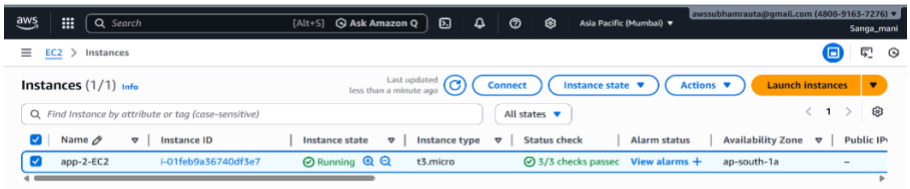
Step	Component	Account	What Was Done	Key Details
1	VPC Creation	Account-1	Created VPC	CIDR: 10.1.0.0/16
		Account-2	Created VPC	CIDR: 192.168.0.0/16
2	Subnet Creation	Account-1	Created public subnet	10.1.1.0/24
		Account-2	Created public subnet	192.168.1.0/24
3	Internet Gateway	Account-1	IGW created & attached	Enabled public access
		Account-2	IGW created & attached	Enabled public access
4	Route Table (Internet)	Both	Added default route	0.0.0.0/0 → IGW
5	**EC2 Launch**	Account-1	EC2 instance launched	Private IP: 10.1.1.51

Step	Component	Account	What Was Done	Key Details
		Account-2	EC2 instance launched	Private IP: 192.168.1.70
6	VPC Peering	Account-1	Peering request created	Cross-account peering
7	Peering Acceptance	Account-2	Accepted peering request	Status: Active
8	Route Table (Peering)	Account-1	Added peering route	192.168.0.0/16 → pcx
		Account-2	Added peering route	10.1.0.0/16 → pcx
9	Subnet Association	Both	Route table associated	Subnet uses peering route
10	Security Groups	Account-1	Allowed ICMP & SSH	Source: 192.168.0.0/16
		Account-2	Allowed ICMP & SSH	Source: 10.1.0.0/16
11	Connectivity Test	Account-1	Ping test	ping 192.168.1.70
12	Port Validation	Account-1	Telnet to SSH port	telnet 192.168.1.70 22
13	Service Check	Both	Verified SSH listening	sshd on port 22

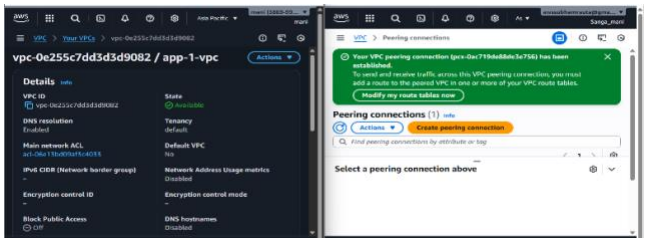
Screenshots

Launch 2 instances: One in own account and 2nd in the different account, **in same region**





Establish the peering connection between two accounts



Access both the servers via **putty**

```
login as: ec2-user
Authenticating with public key "lambda-key"
,     #_
~\_  ####_        Amazon Linux 2023
~~  \_#####\
~~     \###|
~~       \#/ ___    https://aws.amazon.com/linux/amazon-linux-2023
 ~~       V~' '->
  ~~~         /
    ~~._.   _/
       _/ _/
     _/m/'
Last login: Fri Feb  6 14:26:48 2026 from 49.42.189.165
[ec2-user@ip-10-1-1-51 ~]$ ip r
default via 10.1.1.1 dev ens5 proto dhcp src 10.1.1.51 metric 512
10.1.0.2 via 10.1.1.1 dev ens5 proto dhcp src 10.1.1.51 metric 512
10.1.1.0/24 dev ens5 proto kernel scope link src 10.1.1.51 metric 512
10.1.1.1 dev ens5 proto dhcp scope link src 10.1.1.51 metric 512
[ec2-user@ip-10-1-1-51 ~]$
[ec2-user@ip-10-1-1-51 ~]$
```

```
login as: ec2-user
Authenticating with public key "PEERING-key"
,     #_
~\_  ####_        Amazon Linux 2023
~~  \_#####\
~~     \###|
~~       \#/ ___    https://aws.amazon.com/linux/amazon-linux-2023
 ~~       V~' '->
  ~~~         /
    ~~._.   _/
       _/ _/
     _/m/'
Last login: Fri Feb  6 14:33:36 2026 from 49.42.189.165
[ec2-user@ip-192-168-1-70 ~]$ ip r
default via 192.168.1.1 dev ens5 proto dhcp src 192.168.1.70 metric 512
192.168.0.2 via 192.168.1.1 dev ens5 proto dhcp src 192.168.1.70 metric 512
192.168.1.0/24 dev ens5 proto kernel scope link src 192.168.1.70 metric 512
192.168.1.1 dev ens5 proto dhcp scope link src 192.168.1.70 metric 512
[ec2-user@ip-192-168-1-70 ~]$
[ec2-user@ip-192-168-1-70 ~]$
```

Ping: To check whether one server can reach another server.

```
    ~~._.   _/
       _/ _/
     _/m/'
Last login: Fri Feb  6 14:26:48 2026 from 49.42.189.165
[ec2-user@ip-10-1-1-51 ~]$ ip r
default via 10.1.1.1 dev ens5 proto dhcp src 10.1.1.51 metric 512
10.1.0.2 via 10.1.1.1 dev ens5 proto dhcp src 10.1.1.51 metric 512
10.1.1.0/24 dev ens5 proto kernel scope link src 10.1.1.51 metric 512
10.1.1.1 dev ens5 proto dhcp scope link src 10.1.1.51 metric 512
[ec2-user@ip-10-1-1-51 ~]$
[ec2-user@ip-10-1-1-51 ~]$
[ec2-user@ip-10-1-1-51 ~]$ ping 192.168.1.70
PING 192.168.1.70 (192.168.1.70) 56(84) bytes of data.
64 bytes from 192.168.1.70: icmp_seq=1 ttl=127 time=0.466 ms
64 bytes from 192.168.1.70: icmp_seq=2 ttl=127 time=0.197 ms
64 bytes from 192.168.1.70: icmp_seq=3 ttl=127 time=0.191 ms
64 bytes from 192.168.1.70: icmp_seq=4 ttl=127 time=0.180 ms
64 bytes from 192.168.1.70: icmp_seq=5 ttl=127 time=0.193 ms
64 bytes from 192.168.1.70: icmp_seq=6 ttl=127 time=0.184 ms
64 bytes from 192.168.1.70: icmp_seq=7 ttl=127 time=0.204 ms
64 bytes from 192.168.1.70: icmp_seq=8 ttl=127 time=0.195 ms
64 bytes from 192.168.1.70: icmp_seq=9 ttl=127 time=0.191 ms
64 bytes from 192.168.1.70: icmp_seq=10 ttl=127 time=0.207 ms
64 bytes from 192.168.1.70: icmp_seq=11 ttl=127 time=0.185 ms
64 bytes from 192.168.1.70: icmp_seq=12 ttl=127 time=0.198 ms
64 bytes from 192.168.1.70: icmp_seq=13 ttl=127 time=0.227 ms
64 bytes from 192.168.1.70: icmp_seq=14 ttl=127 time=0.187 ms
64 bytes from 192.168.1.70: icmp_seq=15 ttl=127 time=0.186 ms
64 bytes from 192.168.1.70: icmp_seq=16 ttl=127 time=0.221 ms
64 bytes from 192.168.1.70: icmp_seq=17 ttl=127 time=0.188 ms
64 bytes from 192.168.1.70: icmp_seq=18 ttl=127 time=0.210 ms
64 bytes from 192.168.1.70: icmp_seq=19 ttl=127 time=0.188 ms
64 bytes from 192.168.1.70: icmp_seq=20 ttl=127 time=0.514 ms
64 bytes from 192.168.1.70: icmp_seq=21 ttl=127 time=0.190 ms
64 bytes from 192.168.1.70: icmp_seq=22 ttl=127 time=0.188 ms
^C
--- 192.168.1.70 ping statistics ---
22 packets transmitted, 22 received, 0% packet loss, time 21858ms
rtt min/avg/max/mdev = 0.180/0.222/0.514/0.085 ms
```

```
login as: ec2-user
Authenticating with public key "PEERING-key"
,     #_
~\_  ####_        Amazon Linux 2023
~~  \_#####\
~~     \###|
~~       \#/ ___    https://aws.amazon.com/linux/amazon-linux-2023
 ~~       V~' '->
  ~~~         /
    ~~._.   _/
       _/ _/
     _/m/'
Last login: Fri Feb  6 14:33:36 2026 from 49.42.189.165
[ec2-user@ip-192-168-1-70 ~]$ ip r
default via 192.168.1.1 dev ens5 proto dhcp src 192.168.1.70 metric 512
192.168.0.2 via 192.168.1.1 dev ens5 proto dhcp src 192.168.1.70 metric 512
192.168.1.0/24 dev ens5 proto kernel scope link src 192.168.1.70 metric 512
192.168.1.1 dev ens5 proto dhcp scope link src 192.168.1.70 metric 512
[ec2-user@ip-192-168-1-70 ~]$
[ec2-user@ip-192-168-1-70 ~]$
[ec2-user@ip-192-168-1-70 ~]$ ping 10.1.1.51
PING 10.1.1.51 (10.1.1.51) 56(84) bytes of data.
64 bytes from 10.1.1.51: icmp_seq=1 ttl=127 time=0.190 ms
64 bytes from 10.1.1.51: icmp_seq=2 ttl=127 time=0.207 ms
64 bytes from 10.1.1.51: icmp_seq=3 ttl=127 time=0.196 ms
64 bytes from 10.1.1.51: icmp_seq=4 ttl=127 time=0.256 ms
64 bytes from 10.1.1.51: icmp_seq=5 ttl=127 time=0.200 ms
64 bytes from 10.1.1.51: icmp_seq=6 ttl=127 time=0.204 ms
64 bytes from 10.1.1.51: icmp_seq=7 ttl=127 time=0.203 ms
64 bytes from 10.1.1.51: icmp_seq=8 ttl=127 time=0.203 ms
64 bytes from 10.1.1.51: icmp_seq=9 ttl=127 time=0.209 ms
64 bytes from 10.1.1.51: icmp_seq=10 ttl=127 time=0.227 ms
64 bytes from 10.1.1.51: icmp_seq=11 ttl=127 time=0.243 ms
64 bytes from 10.1.1.51: icmp_seq=12 ttl=127 time=0.241 ms
^C
--- 10.1.1.51 ping statistics ---
12 packets transmitted, 12 received, 0% packet loss, time 11478ms
rtt min/avg/max/mdev = 0.190/0.214/0.256/0.020 ms
[ec2-user@ip-192-168-1-70 ~]$
```

Install **telnet:** to check if a port is open on a server.

```
==============================================================================
Installing:
 telnet      x86_64      1:0.17-83.amzn2023.0.2       amazonlinux       64 k

Transaction Summary
==============================================================================
```

```
==============================================================================
Installing:
 telnet      x86_64      1:0.17-83.amzn2023.0.2       amazonlinux       64 k

Transaction Summary
==============================================================================
```

```
==============================================================================


Installed:
  telnet-1:0.17-83.amzn2023.0.2.x86_64

Complete!
[ec2-user@ip-10-1-1-51 ~]$
```

```
==============================================================================

Installed:
  telnet-1:0.17-83.amzn2023.0.2.x86_64

Complete!
[ec2-user@ip-192-168-1-70 ~]$
```

sudo lsof -i :22 is used to check which service is using and listening on SSH port 22. (on 1st account)

```
[ec2-user@ip-10-1-1-51 ~]$ sudo lsof -i :22
COMMAND  PID     USER   FD   TYPE DEVICE SIZE/OFF NODE NAME
sshd    1551     root    3u  IPv4  16051      0t0  TCP *:ssh (LISTEN)
sshd    1551     root    4u  IPv6  16053      0t0  TCP *:ssh (LISTEN)
sshd    4295     root    4u  IPv4  22382      0t0  TCP ip-10-1-1-51.ap-south-
1.compute.internal:ssh->49.42.188.13:59866 (ESTABLISHED)
sshd    4313 ec2-user    4u  IPv4  22382      0t0  TCP ip-10-1-1-51.ap-south-
1.compute.internal:ssh->49.42.188.13:59866 (ESTABLISHED)
sshd    5266     root    4u  IPv4  25804      0t0  TCP ip-10-1-1-51.ap-south-
1.compute.internal:ssh->49.42.189.165:59218 (ESTABLISHED)
sshd    5269 ec2-user    4u  IPv4  25804      0t0  TCP ip-10-1-1-51.ap-south-
1.compute.internal:ssh->49.42.189.165:59218 (ESTABLISHED)
sshd    5551     root    4u  IPv4  26191      0t0  TCP ip-10-1-1-51.ap-south-
1.compute.internal:ssh->49.42.189.165:65349 (ESTABLISHED)
sshd    5559 ec2-user    4u  IPv4  26191      0t0  TCP ip-10-1-1-51.ap-south-
1.compute.internal:ssh->49.42.189.165:65349 (ESTABLISHED)
```

(on 2nd account)

```
[ec2-user@ip-192-168-1-70 ~]$ sudo lsof -i :22
COMMAND  PID     USER   FD   TYPE DEVICE SIZE/OFF NODE NAME
sshd    1573     root    3u  IPv4  17470      0t0  TCP *:ssh (LISTEN)
sshd    1573     root    4u  IPv6  17480      0t0  TCP *:ssh (LISTEN)
sshd    4838     root    4u  IPv4  24848      0t0  TCP ip-192-168-1-70.ap-south-1.compute.internal:ssh->49.42.189.165:50889 (ESTABLISHED)
sshd    4851 ec2-user    4u  IPv4  24848      0t0  TCP ip-192-168-1-70.ap-south-1.compute.internal:ssh->49.42.189.165:50889 (ESTABLISHED)
[ec2-user@ip-192-168-1-70 ~]$
```

sudo systemctl status sshd is used to check whether the SSH service is running on the server.

```
[ec2-user@ip-10-1-1-51 ~]$ sudo systemctl status sshd
● sshd.service - OpenSSH server daemon
   Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: d
   Active: active (running) since Fri 2026-02-06 12:16:02 UTC; 2h 5min ago
     Docs: man:sshd(8)
           man:sshd_config(5)
 Main PID: 1551 (sshd)
    Tasks: 1 (limit: 1067)
   Memory: 9.4M
      CPU: 2min
   CGroup: /system.slice/sshd.service
           └─1551 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"
```

```
[ec2-user@ip-192-168-1-70 ~]$ sudo systemctl status sshd
● sshd.service - OpenSSH server daemon
   Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: d
   Active: active (running) since Fri 2026-02-06 12:36:29 UTC; 2h 22min ago
     Docs: man:sshd(8)
           man:sshd_config(5)
 Main PID: 1573 (sshd)
    Tasks: 1 (limit: 1067)
   Memory: 5.4M
      CPU: 232ms
   CGroup: /system.slice/sshd.service
           └─1573 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"
```